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United States Patent	12404687
Kind Code	B2
Date of Patent	September 02, 2025
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Fall arrest roof mount anchor

Abstract

A fall arrest roof mount anchor is designed to secure construction and maintenance personnel working at heights, utilising harness and lanyard fall protection equipment. The roof mount anchor comprises a mounting plate having a uniform cross-section defining a raised central portion and side portions for roof attachment. The raised central portion has a swivel eye anchor point for attachment of conventional height safety harness hardware.

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Appl. No.:	18/348679
Filed:	July 07, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230349181 A1	Nov. 02, 2023

Foreign Application Priority Data

AU	2021902563	Aug. 17, 2021
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Related U.S. Application Data

continuation parent-doc US 18001924 PENDING WO PCT/AU2022/050890 20220815 child-doc US 18348679

Publication Classification

Int. Cl.: E04G21/32 (20060101); A62B35/00 (20060101); A62B35/04 (20060101)

U.S. Cl.:

CPC E04G21/328 (20130101); A62B35/0068 (20130101); A62B35/04 (20130101); E04G21/3214 (20130101); E04G21/3276 (20130101);

Field of Classification Search

CPC: E04G (21/328); E04G (21/3276); E04G (21/3214); E04G (21/3261); E04G (21/329); E04G (3/26); A62B (35/0068); A62B (35/04); A62B (35/0037); Y02E (10/47); E04D (13/12)

USPC: 52/98; 52/573.1; 52/749.12; 52/23; 52/24; 52/698; 52/704; 52/710; 248/237; 182/3; 182/45

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Background/Summary

FIELD OF THE INVENTION

(1) This invention relates generally to a fall arrest roof mount anchor.

BACKGROUND OF THE INVENTION

(2) Fall arrest roof mount anchors are used for industry applications involving high fall risk, to secure harnessed construction and maintenance personnel working at heights on buildings and structures.

(3) These height safety anchors may comprise a formed aluminium extrusion mounting plate which is permanently affixed to a roof profile surface, and which exposes a swivel eye anchor point for attachment of a carabiner, lanyard and other conventional harness hardware.

(4) The present invention seeks to provide a fall arrest roof mount anchor which will overcome or substantially ameliorate at least some of the deficiencies of the prior art, or at least to provide an alternative.

(5) It is to be understood that, if any prior art information is referred to herein, such reference does not constitute an admission that the information forms part of the common general knowledge in the art, either in Australia or any other country.

SUMMARY OF THE DISCLOSURE

(6) There is provided herein a fall arrest roof mount anchor ideally suited to construction and maintenance personnel working at heights, utilising a harness and lanyard fall protection

equipment.

(7) The fall arrest roof mount anchor has a mounting plate having a uniform cross-section defining a raised central portion and side portions for roof attachment.

(8) The raised central portion has a swivel eye anchor point.

(9) The mounting plate further comprises channel wall portions projecting from an under surface of the raised central portion thereby defining an integrally formed channel therebetween, which runs longitudinally along the under surface of the raised central portion and coincides with the swivel eye anchor point.

(10) The uniform cross-section and the under channel enhance shear force resilience and flexural stiffness to lateral applied strain, especially when applied along a longitudinal axis of the channel. The under channel coincides with the swivel eye anchor point, thereby maximising the shear force resilience conferred by the under channel at the point of fall arrest loading.

(11) As such, the present roof mount anchor may be structurally resilient yet lightweight. For example, an aluminium extrusion embodiment of the present anchor may be rated to 15 kN yet only weigh 800 g.

(12) The under channel may further be configured to conveniently capture a locknut of an anchor bolt whilst yet allowing a distal end of the anchor bolt to extend therebetween.

(13) The roof attachment side portions may comprise a plurality of rivet fastener holes which collocate along a length of the horizontal axis and are spaced rows in across the horizontal axis. Spacing of the rivet fastener holes along the rows accommodate specific crest-to-crest dimensions of various roof profile surfaces, wherein the rivet fasteners are spaced apart in at least two intervals of 14 mm, 28 mm, 42 mm, 230 mm, 245 mm and 274 mm from an edge of each roof attachment side portion of the mounting plate.

(14) As such, the connection of the side portions of the mounting plate across two roof profile ribs dissipates the load over a greater area and enhances fall arrest rated performance. Furthermore, installation across two ribs prevents or reduces potential for roof sheet delamination, which could occur when a roof mount is fixed with rivets to one roof sheet only.

(15) The roof mount anchor may incorporate a unique energy absorbing swivel eye lanyard attachment point, eliminating carabiner out and providing uniform load distribution to the mounting plate in a fall arrest situation. The increased energy absorption properties and ability to be installed across two roof sheets allows fixture to lighter structures and roof decks.

(16) Specifically, the mounting plate may comprise a matrix of elongate stress relief cut-outs surrounding the swivel eye anchor point. The stress relief cut-outs may arc concentrically around the swivel eye anchor point, and the ends of radially adjacent stress relief cut-outs may overlap.

(17) This unique energy deforming matrix arrangement allows the mounting plate to flex around the swivel eye anchor point, thereby dissipating energy in a shock load situation without compromising the structural integrity of the remainder of the mounting plate.

(18) The swivel eye anchor point may comprise a lanyard connector attached to the centre of the raised central portion arch apex. The lanyard connector may have an anchor connection aperture being elongate along a strain absorbing axis through proximal and distal ends of the lanyard connector, and wherein the anchor connection aperture has a curved frail web across the strain absorbing axis and wherein the frail web is orientated with an apex towards the anchor bolt.

(19) The curved frail web and the inverse orientation thereof are designed to absorb energy and reduce shock impact on the user, whilst also providing a visual indicator of abnormal loading (such as in excess of 7 kN), which is useful when undertaking routine inspection of the fall arrest system. The inverse orientation in particular provides improved visual indication of gradual energy absorption deformation, as compared to the opposite orientation.

(20) Other aspects of the invention are also disclosed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Notwithstanding any other forms which may fall within the scope of the present invention, preferred embodiments of the disclosure will now be described, by way of example only, with reference to the accompanying drawings in which:
- (2) FIG. 1 shows a perspective view of a fall arrest roof mount anchor in accordance with an embodiment;
- (3) FIG. 2 shows a uniform vertical cross-section view of the roof mount anchor;
- (4) FIG. 3 shows a top plan view of the roof mount anchor;
- (5) FIG. 4 shows a perspective view of a lanyard connector of the roof mount anchor;
- (6) FIG. 5 shows a detail view of an anchor connection aperture of the lanyard connector in accordance with an embodiment;
- (7) FIG. 6 shows a top plan view of the lanyard connector having exemplary dimensions;
- (8) FIG. 7 shows a symmetrical left and right side view of the lanyard connector having exemplary dimensions;
- (9) FIG. 8 shows a raised distal end view of the lanyard connector having exemplary dimensions;
- (10) FIG. 9 shows a uniform vertical cross-section view of a mounting plate of the roof mount anchor having exemplary dimensions;
- (11) FIG. 10 shows a detail view of the anchor connection aperture of FIG. 5 having exemplary dimensions;
- (12) FIG. 11 shows a top plan view of the mounting plate of the roof mount anchor having exemplary dimensions; and
- (13) FIG. 12 shows a lanyard connector according to a further embodiment.

DESCRIPTION OF EMBODIMENTS

- (14) With reference to FIGS. 1 and 2, a fall arrest roof mount anchor **100** has a mounting plate **101** having a uniform cross-section along a length of a horizontal axis **102** thereof. The cross-section defines a raised central portion **103** and roof attachment side portions **104**.
- (15) The mounting plate **101** may preferably be metallic and further preferably aluminium and manufactured using an extrusion process.
- (16) The mounting plate preferably curves smoothly between the side portions **104** and the raised central portion **103**, thereby eliminating sharp transitions and/or planar surfaces which could compromise flexural stiffness. The roof attachment side portions **104** are coplanar so as to lie flat on the roof surface or cross ribs thereof. Exemplary dimensions are given in FIG. 9 wherein the mounting plate **101** may have a low profile height of less than 30 mm, such as approximately 25 mm which to maintain shear force resilience and flexural stiffness.
- (17) The mounting plate **101** may comprise an integrally formed channel **105** running longitudinally along an under-surface of the raised central portion **103**. The channel **105** confers flexural stiffness to the raised central portion **103**. As further illustrated in FIG. 1, the channel **105** may capture a locknut **107** of an anchor bolt **108**.
- (18) Specifically, the mounting plate **101** may further comprise channel wall portions **106** (which are independent of the raised central portion **103** and side portions **104**) which define the channel **105**. The channel wall portions **106** may be parallel thereby defining parallel inner surfaces, and spaced to accommodate a minimum diameter of the locknut **107** and thereby to non-rotatably capture the locknut **107** within the channel **105**.
- (19) Furthermore, the channel **105** may accommodate a central anchor point connection hole **112** along the length thereof, allowing a distal end of the anchor bolt **108** to protrude therethrough as shown in FIG. 2. Specifically, the channel **105** may further comprise inwardly projecting flange portions **109** from respective distal ends of the channel wall portions **106**. The flange portions **109**

may support the locknut **107** within the channel **105** whilst allowing a distal end of the anchor bolt **108** to extend therebetween.

(20) The mounting plate **101** may be configured for installation along adjacent ribs of a roof profile surface, and/or wherein the horizontal axis **102** aligns along the pitch of the roof, or at least orientates towards an edge of the roof.

(21) With reference to FIG. 3, the roof attachment side portions **104** may comprise a plurality of rivet fastener holes **124**. The rivet fastener holes **124** may run in spaced rows **125** across the horizontal axis **102**. The rows **125** may have rivet fastener hole spacings suited for specific roof profile crest-to-crest widths. With reference to FIG. 1, the rivet fastener holes **124** in the rows **125** may be spaced apart in at least two intervals of 14 mm, 28 mm, 42 mm, 230 mm, 245 mm and 274 mm, from an edge of a roof attachment side **104** of the mounting plate **101**.

(22) In the embodiment shown in FIG. 3, the mounting plate **101** comprises three rows **125**, however in embodiments, a mounting plate **101** having a different number of rows **125** is contemplated, including an embodiment wherein the mounting plate **101** has four rows **125**.

(23) The roof mount anchor **100** may comprise mounting pads **126** preferably of rubber material, which engage under each roof attachment side **104** of the mounting plate **101**. The mounting pads **126** may comprise a plurality of fastener holes **127** which collocate with the rivet fastener holes **124** of the mounting plate **101**.

(24) As shown in FIG. 3, the mounting plate **101** may further splay longitudinally along the horizontal axis **102** towards both ends thereof.

(25) The roof mount anchor **100** may further comprise a lanyard connector **110** attached to an apex of the raised central portion **103**. With reference to FIG. 4, the lanyard connector **110** may define an anchor connection aperture **111** for swivel engagement by the anchor bolt **108** at a proximal end **113** thereof. The mounting plate **101** may accordingly comprise a central anchor point connection hole **112** for the anchor bolt **108**.

(26) The lanyard connector **110** may define an eyelet **114** towards a distal end **115** thereof, through which a carabiner or the like may attach a lanyard for tethering a user thereto.

(27) The lanyard connector **110** may comprise a swivel eye anchor plate **116** which bends upwardly towards the distal end **115**. The eyelet **114** may transition across the upwardly bent distal end **115**.

(28) With reference to FIG. 5, the anchor connection aperture **111** may be elongate along a strain absorbing axis **117** through the proximal end **113** and distal end **115** of the lanyard connector **110**. The lanyard connector **110** may have an integrally formed primary frail web **118** across the strain absorbing axis **117**. When strain is applied to the eyelet **114** of the lanyard connector **110**, a resultant dynamic force is applied by the anchor bolt **108** against the primary frail web **118**.

(29) The anchor connection aperture **111** may define a profile open an anchor bolt accommodation portion **119** which accommodates the anchor bolt **108** therethrough. The primary frail web **118** is preferably curved across the anchor connection aperture **111**. The primary frail web **118** is further preferably inversely orientated so that an apex **120** thereof bears oppositely against the anchor bolt **108**. This arrangement is designed both to absorb energy and reduce shock impact on the user, whilst also providing visual indication of abnormal loading (such as in excess of 7 kN), which is useful when undertaking routine inspection of the fall arrest system

(30) The primary frail web **118** may have an integral seating profile **121** which supports the anchor bolt **108** centrally against the primary frail web apex **120**. The seating profile **121** may have an opposite curvature to that of the primary frail web **118**.

(31) FIG. 12 shows an embodiment wherein the seating profile **121**, extends right to the edges of the anchor bolt cut-out **119** of the anchor connection aperture **111**, thereby not only seating the anchor bolt **108** centrally but also reinforcing the primary frail web **118**. As such, according to this embodiment, the seating profile **121** may itself deform or fracture when the primary frail web **118** deforms under excessive load.

(32) The primary frail web **118** may have a thickness so as to visibly deform when a dynamic force

of approximately 7 kN is applied to the lanyard connector **110**. In this regard, with reference to FIGS. 7 and **10**, the lanyard connector **110** may be stainless steel comprising a thickness of approximately 3 mm, and the primary frail web **118** width be approximately 1.5 mm. The primary frail web **118** may further absorb dynamic shock forces when deforming.

(33) The anchor connection aperture **111** may further comprise an integrally formed secondary frail web **122** across the strain absorbing axis **117**. The secondary frail web **122** width may be narrower than the primary frail web **118** width, such as being approximately 1 mm as shown in FIG. **10**.

(34) With reference to FIGS. **1** and **3**, the mounting plate **101** may comprise a matrix **130** of elongate stress relief cut-outs **123** surrounding the swivel eye anchor point. Each stress relief cut-out **123** may arc concentrically around the swivel eye anchor point.

(35) Single and pairs of stress relief cut-outs **123** may locate symmetrically opposite the swivel eye anchor point connection hole **112**, and may be arranged at different radial offsets. Furthermore, the stress relief cut-outs **123** arranged at different radial offsets from the swivel eye anchor point may overlap concentrically.

(36) The foregoing description, for purposes of explanation, used specific nomenclature to provide a thorough understanding of the invention. However, it will be apparent to one skilled in the art that specific details are not required in order to practise the invention. Thus, the foregoing descriptions of specific embodiments of the invention are presented for purposes of illustration and description. They are not intended to be exhaustive or to limit the invention to the precise forms disclosed, as obviously many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the invention and its practical applications, thereby enabling others skilled in the art to best utilise the invention and various embodiments, with various modifications as are suited to the particular use contemplated. It is intended that the following claims and their equivalents define the scope of the invention.

Claims

1. A lanyard connector having an anchor connection aperture being elongate along a strain absorbing axis through proximal and distal ends of the lanyard connector, and wherein the anchor connection aperture has an anchor bolt accommodation portion and a curved frail web adjacent to the anchor bolt accommodation portion and across the strain absorbing axis, and wherein the curved frail web is orientated with an apex thereof towards the anchor bolt accommodation portion.
 2. The lanyard connector as claimed in claim 1, wherein the curved frail web has an integral seating profile extending from an apex thereof, which holds an anchor bolt within the anchor bolt accommodation portion centrally against the curved frail web apex.
 3. The lanyard connector as claimed in claim 2, wherein the seating profile has an opposite curvature to that of the curved frail web.
 4. The lanyard connector as claimed in claim 1, wherein the anchor connection aperture has an integrally formed secondary frail web across the strain absorbing axis.
 5. The lanyard connector as claimed in claim 4, wherein the secondary frail web width is narrower than the curved frail web width.
 6. The lanyard connector as claimed in claim 1, wherein the curved frail web is configured to deform visibly when a dynamic shock force of approximately 7 kN is applied to the lanyard connector.
 7. The lanyard connector as claimed in claim 6, wherein the lanyard connector is stainless steel having a thickness of approximately 3 mm, and the curved frail web having a width of approximately 1.5 mm.
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